

Experiment Design and Rule Catalog

Design

The experiment consists of six sessions with five blocks each, one per rule family: Arithmetic, Attraction, Recoloring, Expansion, and Occlusion.

Each block contains one rule family and one of two contexts (Figure 1):

- **Previous rule (Inference):** compare the current rule with the rule from the previous trial.
- **First/Memorized rule (Application):** compare the current rule with the rule introduced at the start of the block.

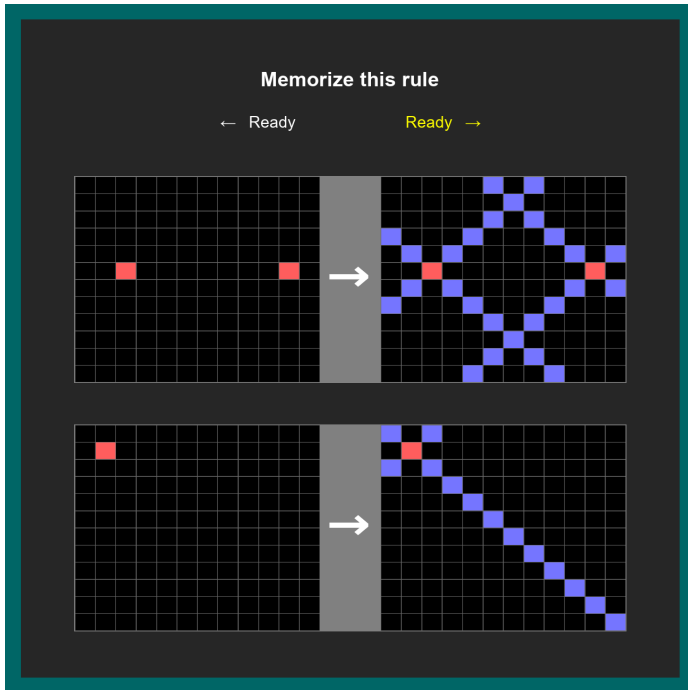
Each trial presents two examples of the same hidden rule. A block contains one initial rule trial followed by eight same/different decisions, balanced to four *same* and four *different* trials with a maximum run length of three identical responses.

Contexts alternate within sessions and the starting context alternates across sessions. Rule families are assigned such that each family occurs equally often in both contexts across the six sessions.

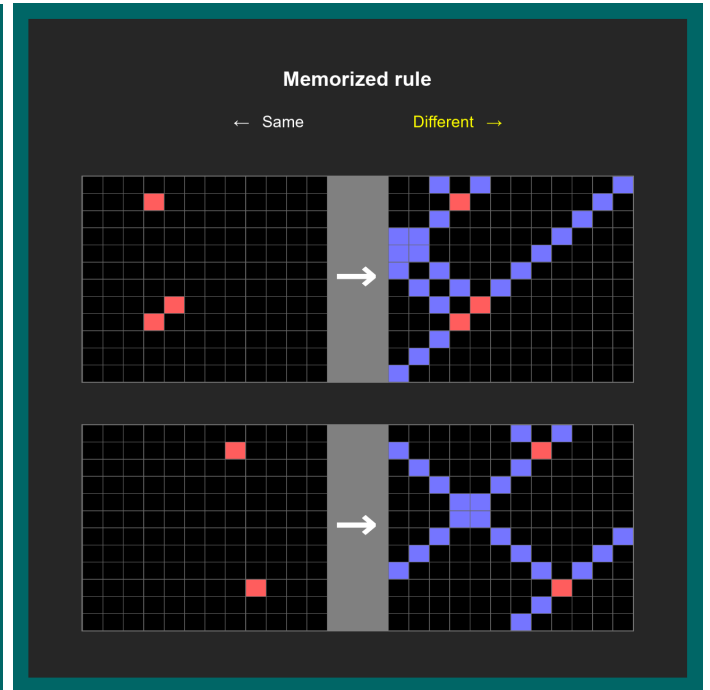
Frame color is counterbalanced across participants: Variant A (odd participant numbers) uses yellow for Inference and cyan for Application; Variant B (even participant numbers) reverses this mapping.

Trials last 10 s and each block is preceded by 10 s fixation. Including the final fixation and 10 s of dummy scans (e.g. $TR = 2$ s), each session requires 520 s (8 min 40 s; 260 volumes).

Application / Memorized rule

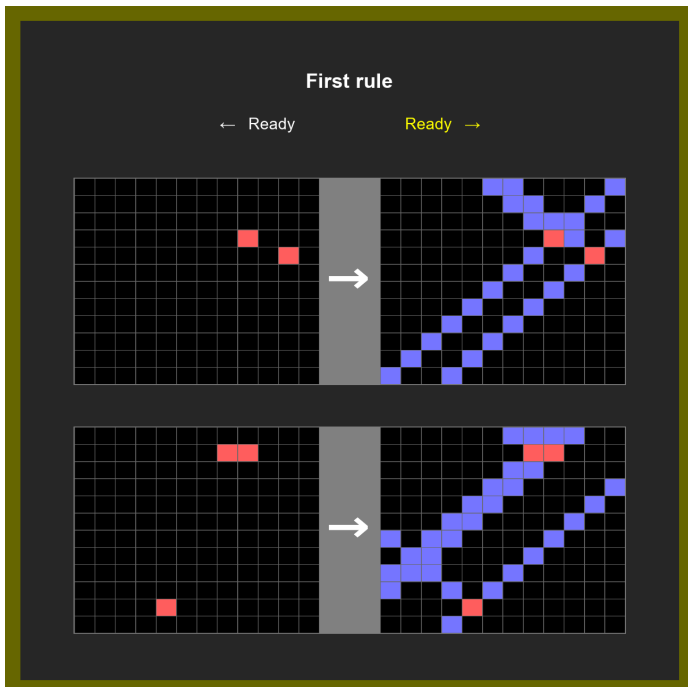


Initial trial

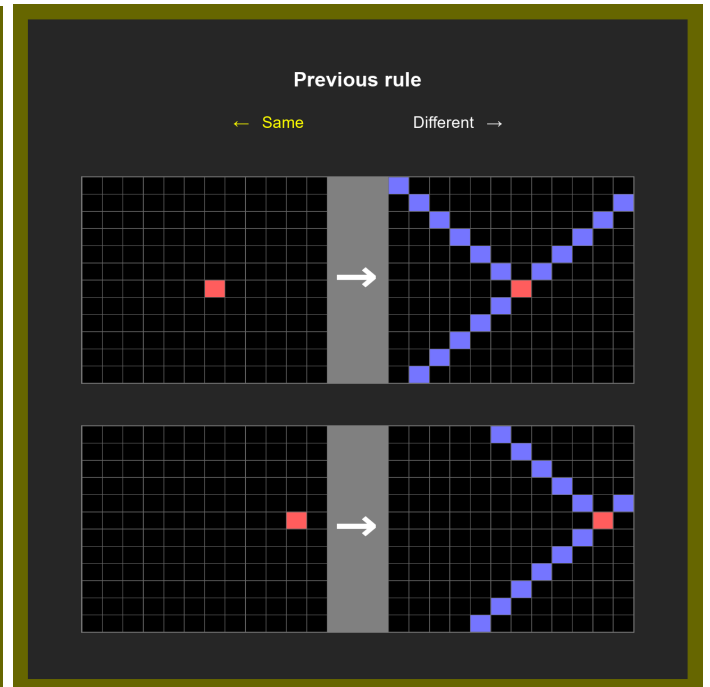


Decision trial

Inference / Previous rule



Initial trial



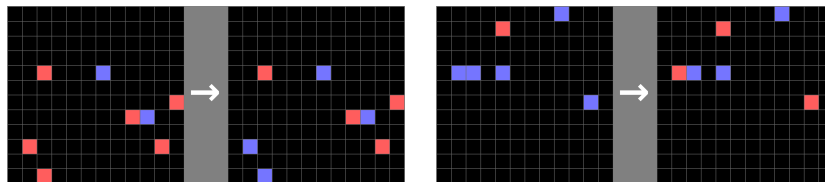
Decision trial

Figure 1: Example screens for the two experimental contexts (Variant A).

Arithmetic

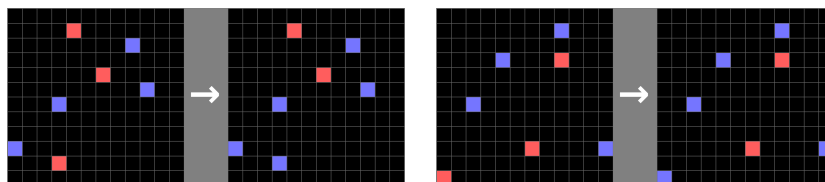
- Equalize colors

Event: recoloring Condition: color, counting Stimulus: dots



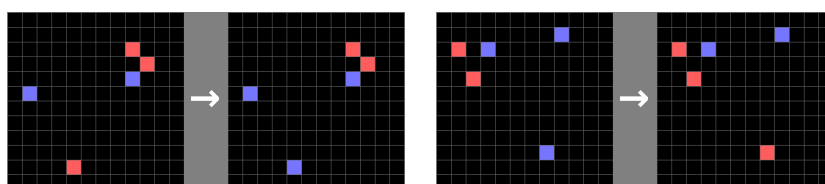
- Increment majority color

Event: recoloring Condition: color, counting Stimulus: dots



- Increment minority color

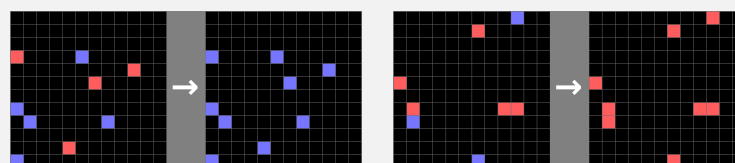
Event: recoloring Condition: color, counting Stimulus: dots



- Majority takeover

Excluded

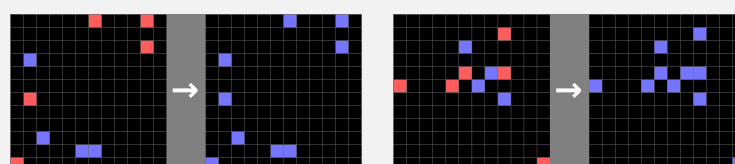
Event: recoloring Condition: color, counting Stimulus: dots



- Minority takeover

Excluded

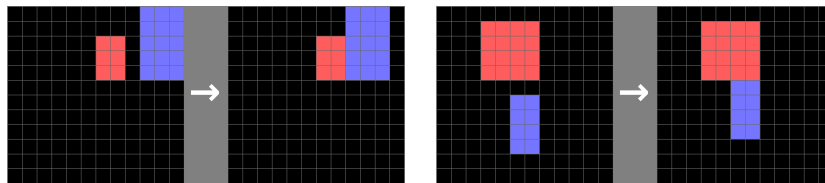
Event: recoloring Condition: color, counting Stimulus: dots



Attraction

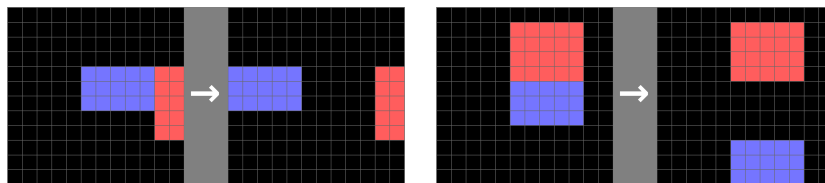
- Color attraction

Event: attraction Condition: color, movement Stimulus: big_blocks



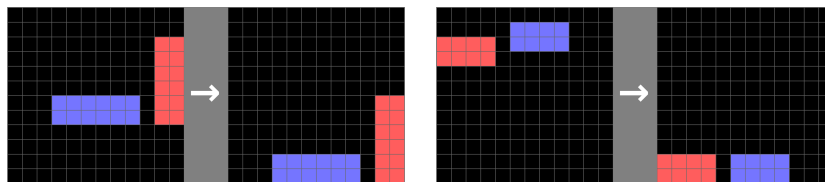
- Color repulsion

Event: repulsion Condition: color, movement Stimulus: big_blocks



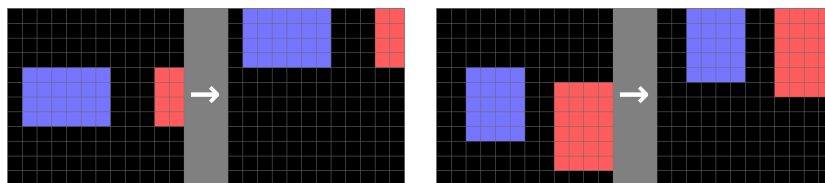
- Falling blocks

Event: falling Condition: movement Stimulus: big_blocks



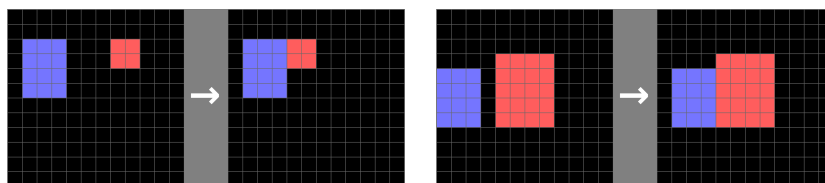
- Floating blocks

Event: floating Condition: movement Stimulus: big_blocks



- Size attraction

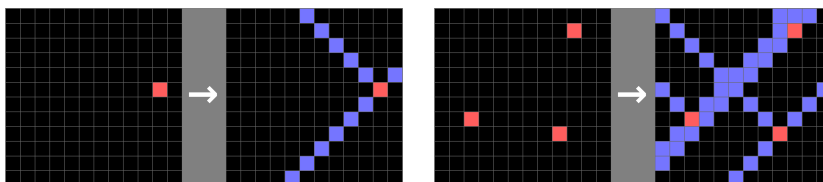
Event: attraction Condition: shape, movement Stimulus: big_blocks



Expansion

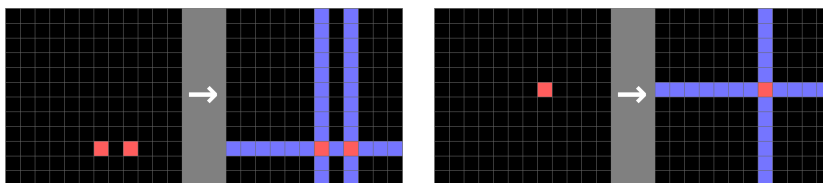
- 3 arm star ray

Event: expansion Condition: shape Stimulus: ray_expansion



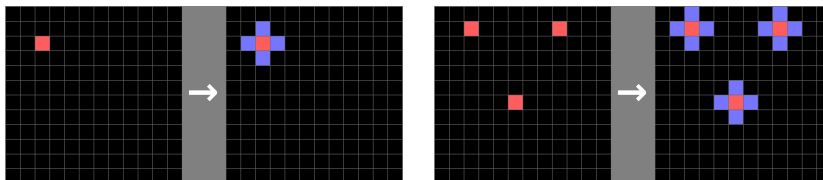
- Plus ray

Event: expansion Condition: shape Stimulus: ray_expansion



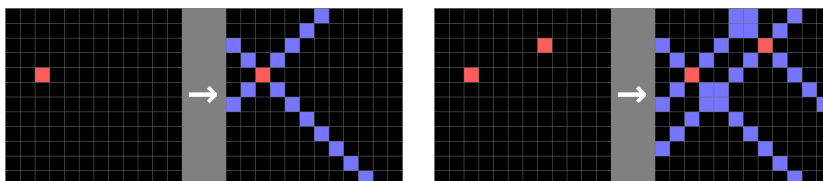
- Plus step

Event: expansion Condition: shape Stimulus: step_expansion



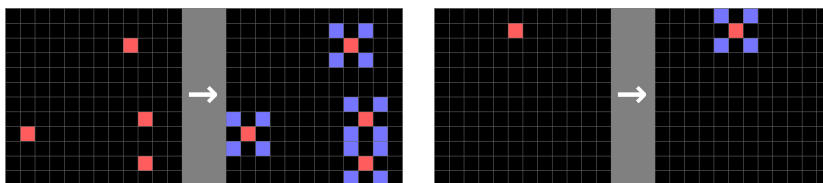
- Star ray

Event: expansion Condition: shape Stimulus: ray_expansion



- Star step

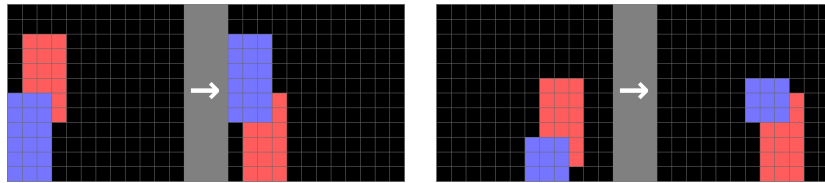
Event: expansion Condition: shape Stimulus: step_expansion



Occlusion

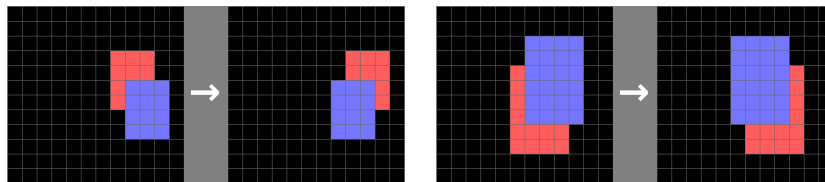
- Mirror x

Event: mirror Condition: shape Stimulus: occluding_blocks



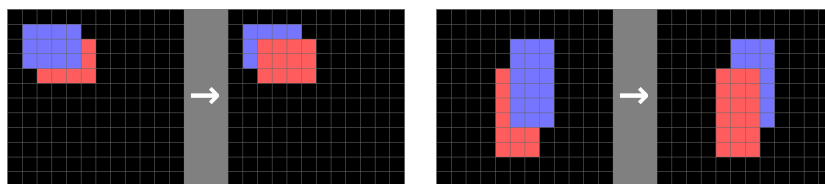
- Mirror y

Event: mirror Condition: shape Stimulus: occluding_blocks



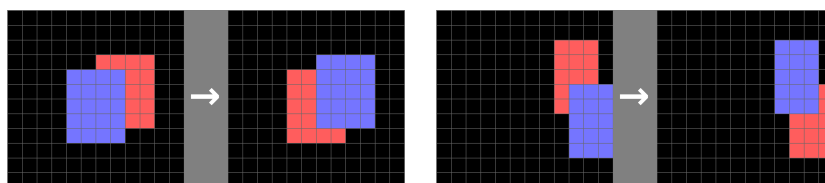
- Occlusion reversal

Event: occlusion_reversal Condition: shape Stimulus: occluding_blocks



- Rotate 180

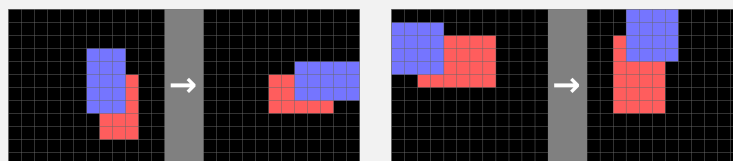
Event: rotation Condition: shape Stimulus: occluding_blocks



- Rotate 90

Excluded

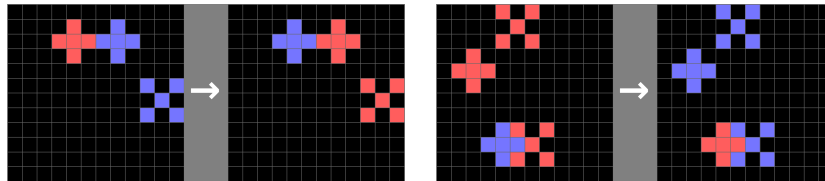
Event: rotation Condition: shape Stimulus: occluding_blocks



Recolor

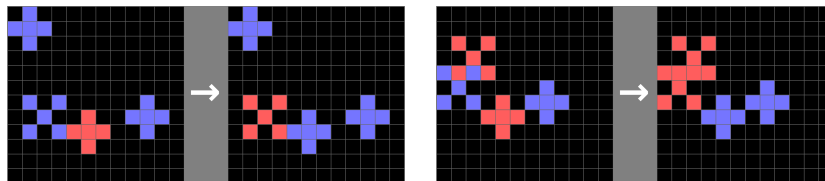
- Color inversion

Event: recoloring Condition: color Stimulus: cross_plus



- Shape color mapping

Event: recoloring Condition: shape Stimulus: cross_plus



- Touching edges recolor

Event: recoloring Condition: shape, neighbor Stimulus: cross_plus

